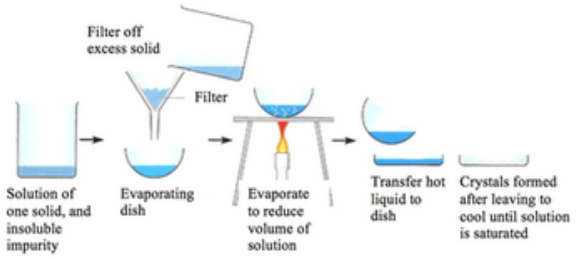
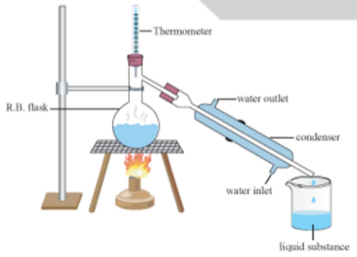
- Crystals are filtered from the solution, washed with distilled water, and dried between the filter paper.



1.6. Simple Distillation

Used to separate a solvent from a solution

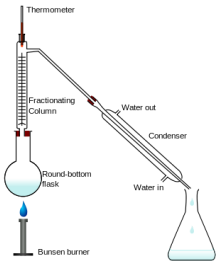
- The impure liquid is heated in a round bottom flask
- When it boils, the steam rises into the attached condenser
- Condenser cools the steam to a pure liquid, and it drops into the beaker
- A thermometer is placed to ensure the **highest boiling point is not exceeded.**



1.8. Fractional Distillation

Used to separate miscible liquids

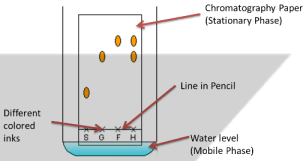
- Mixture is heated
- Substances, due to their different boiling points, rise in different fractions.
- A mixture of gases condenses on the beads in the fractional column.
- The beads are heated to the boiling point of the lowest substance so that the substance being removed cannot condense on the beads.
- The other substances continue to condense and will drip back into the flask.
- The beaker can be changed after every fraction



1.9. Chromatography

Used to separate substances in a solvent with different solubilities.

1. Drop the substance onto the start line (pencil) drawn on chromatography paper.
2. Paper is placed in a beaker with solvent; the paper must touch the surface of the solvent (**water or ethanol**) while the line must be above the liquid.
3. The solvent travels up the paper.
4. Different solubilities lead to different travel rates (**high solubility -> high travel rate**)
   - The stationary phase is the material on which separation takes place
   - The mobile phase is the mixture you want to separate, dissolved in a solvent.



- Interpreting simple chromatograms:

  Chromatograms are the visual outputs on the chromatography paper

  - **Number of rings/dots = number of substances**
  - If two dots travel the same distance up the paper, they are the **same substance.**
  - The pure substance only gives **one spot.**
- Retention Value:

  Used to identify a substance, calculated by the formula:

$$\text{Rf Value} = \frac{\text{Distance moved by substance}}{\text{Distance moved by solvent}}$$

Locating Agents

- Used to make colourless chromatograms visible
  - Dry paper in the oven
  - Spray it with locating agent
  - Heat it for 10 minutes in the oven

1.10. Separation Techniques

- This can be done by dissolving one in an appropriate solvent
- Then filter one and extract the other from the solution by evaporation
- If one solid is magnetic, it can use a magnet, e.g. sand and iron fillings

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